SelectNorm: A Disagreement-Factorized Analysis of Per-Input Normalization for Arabizi with Frozen Encoders

[AUTHOR NAME(S)][1][[ORCID]]

[AFFILIATION, ADDRESS]
[E-MAIL]

Abstract. A classifier trained on Arabic-script text can be applied to Latin-script Arabizi directly or after transliteration, and the better choice varies from input to input. We study whether this choice can be learned per input with frozen encoders and linear heads, and whether it transfers from Tunisian to Algerian and Moroccan Arabizi. For binary labels the expected benefit of normalising factorises into the probability that the views disagree and the margin by which the normalised view then wins, so that disagreement detection, conditional advantage estimation and the cost-aware decision can be measured separately. With four frozen encoders, normalising improves macro-F1 by 2–19 points on Tunisian and Algerian test sets, and on Moroccan text its sign depends on the encoder. Learned per-input routing does not consistently exceed the better single view or strong baselines; against a direct-gain router every paired 95% interval lies within ±0.7 points. Routing reads the first pass before deciding, so it costs more than either single view. Disagreements are moderately predictable (AUC 0.62–0.74), whereas the winning view is weakly predictable from decision-point features (AUC at most 0.71; point estimates below 0.5 in two cases), also when a linear estimator is fitted on target labels. The usable signal is mostly set-level, which about a hundred labels recover.

Keywords: Arabizi · Transliteration · Selective normalisation · Learning to defer · Frozen encoders · Dialect transfer

1 Introduction

North African social-media text is written in Arabic script, in Latin-script Arabizi with digits standing for Arabic consonants (*3* = ع, *7* = ح), and with French segments inside the same sentence. A classifier trained on Arabic-script data can be applied to Arabizi as it is or after transliteration, which helps when it recovers known words and harms when it garbles French segments, negation, names or another dialect's spelling. The choice is usually made once per dataset [25,23,3], although the better choice varies from input to input.

We ask whether a frozen classifier can learn, without labels at inference time, *for which inputs* normalisation pays off, and whether that decision transfers from Tunisian to Algerian and Moroccan Arabizi. Per-input benefit predictors exist: post-hoc deferral rules estimate the gain of a second stage [18], and loss predictors select test-time

transformations [20]. They do not address how such a rule moves to a new dialect with scarce labels, nor which part of it fails.

With binary labels, the expected benefit of normalising equals the probability that the two views disagree times the margin by which the normalised view wins when they do (Eq. (3)). The first factor needs no labels; the second needs labels only on disagreements. The identity is elementary and underlies query-by-committee [30]; we use it as an analysis device that separates disagreement detection, conditional advantage estimation and a cost-aware decision, so that each can be measured. We ask whether per-input selection beats dataset-level choices and strong routers (RQ1) across corpora and dialects (RQ2), what it costs (RQ3), whether normalised predictions agree with predictions on human transliterations (RQ4), which sub-problem limits it (RQ5), and how much a few target labels help (RQ6). Our contributions are:

- a disagreement-factorised analysis of selective normalisation, with a cost-aware Bayes rule that separates sunk from avoidable cost, built from standard tools (Sects. 3–4);
- evidence, for four frozen encoders and three Maghrebi varieties, on how far the *conditional advantage* of the normalised view can be predicted from decision-point features (Sect. 6.3);
- a pre-specified evaluation with timing-checked costs, corrected paired tests and mechanism intervals (Sects. 5–6);
- a conservative Arabizi normaliser with safeguards (not independently validated).

No learned router, SelectNorm included, consistently beats the better single view, and routing costs more than either view; disagreements are moderately predictable, but the winning view is only weakly predictable from the features and estimators tested, even in domain.

2 Related Work

Adaptive selection and deferral. Selective classification abstains under low confidence [12,13]; learning to defer trains a classifier and a deferral rule jointly [24]. Closest to our setting are post-hoc cascade rules: Jitkrittum et al. [18] show that confidence-based deferral is optimal only when the second stage's errors are uniform and estimate the gain of deferring directly, and Gupta et al. [16] learn deferral from token-level uncertainty. LLM routers choose a model from preference data [27] or from predicted weak–strong disagreement [26]. These methods choose between *models*; SelectNorm keeps one frozen model, chooses between two *views* of the input and factorises the gain.

Input transformation and multilingual processing. Kim et al. [20] predict the loss of each test-time augmentation (our B4b). Kang et al. [19] train an LLM to decide whether to translate a query before reasoning. Translate-test is usually chosen per task rather than per input [3], and transliterating unseen scripts can help multilingual encoders [25,23]. None studies transfer of the selection rule across dialects.

Arabizi and dialect normalisation. Arabizi has been converted to Arabic script with rules, lexicons and sequence models [6,31]. Dialect normalisation maps dialectal text to a standard variety with rules and LLMs [7] or with dialect identification [2]; DialUp

adapts models to dialect continua and dialect data to models [4], and North African dialects have been modelled from standard languages [32]. These approaches transform all inputs.

Disagreement-based analysis. Agreement between predictors has been used to estimate accuracy on unlabelled data [28,11], and querying labels where predictors disagree is the principle of query-by-committee [30].

3 Problem Formulation

Setting. Let x be a sentence with binary label $y \in \{0, 1\}$, T a deterministic normaliser from Latin-script Arabizi to Arabic script, and f_θ a frozen classifier with posterior $p_a = f_\theta(v_a)$ on view $v_0 = x$ or $v_1 = T(x)$ and decision $\hat{y}_a = \mathbf{1}[p_a > 1/2]$. A routing policy $g : \mathcal{Z} \to \{0, 1\}$ returns $\hat{y}(x) = \hat{y}_{g(z)}$, where $z = z(x)$ collects the features available at the *decision point*: after $f_\theta(x)$ and $T(x)$ have been computed and before $f_\theta(T(x))$ is run. Let $e_a = \mathbf{1}[\hat{y}_a \neq y]$. Inference costs, relative to the mean cost of $f_\theta(x)$, are $c_0(x)$ for the first pass, $c_T(x)$ for the normaliser and $c_1(x)$ for the second pass $f_\theta(T(x))$. Because z contains p_0 and normaliser metadata, $c_0 + c_T$ is *sunk* when g is decided; only c_1 is *avoidable*.

Objective and Bayes rule. For a trade-off $\lambda \geq 0$ we seek

$$\min_g \ \mathbb{E}\big[e_{g(z)} + \lambda \left(c_0 + c_T + c_1\, g(z)\right)\big]. \tag{1}$$

The sunk term does not depend on g, so conditioning on z gives the minimiser

$$g^\star(z) = \mathbf{1}\big[B(z) > \lambda\, \bar{c}(z)\big], \qquad B(z) = \mathbb{E}[e_0 - e_1 \mid z], \quad \bar{c}(z) = \mathbb{E}[c_1 \mid z], \tag{2}$$

where B is the expected *benefit* of normalising and $\bar{c}$ the expected *avoidable* cost (proof in Appendix A). A router that reads p_0 and m costs at least $c_0 + c_T$, whereas never normalising costs c_0 and always normalising $c_T + c_1$: such routing saves compute only relative to running both views. Post-hoc deferral rules [18] and loss predictors [20] estimate B directly.

Disagreement factorisation (elementary). Let $D = \mathbf{1}[\hat{y}_0 \neq \hat{y}_1]$ and $W = \mathbf{1}[\hat{y}_1 = y]$. Because $e_0 = e_1$ whenever $D = 0$, and exactly one view is correct whenever $D = 1$,

$$B(z) = \pi(z)\,\big(2\rho(z) - 1\big), \qquad \pi(z) = P(D{=}1 \mid z), \quad \rho(z) = P(W{=}1 \mid D{=}1, z). \tag{3}$$

(With K classes, $B = \pi(\rho_1 - \rho_0)$.) The identity is not a theoretical contribution; its factors have different data requirements: π depends only on the two views' predictions and can be estimated on *unlabelled* text, whereas ρ needs labels, but only on disagreements. Equations (2) and (3) split selective normalisation into *disagreement detection* (π), *conditional advantage estimation* (ρ) and a *cost-aware decision* ($\lambda\bar{c}$). Both estimators minimise the log-loss on the policy split ($\hat{\rho}$ on disagreements only); the plug-in rule is $\hat{g} = \mathbf{1}[\hat{\pi}(2\hat{\rho} - 1) > \lambda\bar{c}]$. Two consequences follow. (i) At $\lambda = 0$ and $\hat{\pi} > 0$, $\hat{g} = \mathbf{1}[\hat{\rho} > 1/2]$: re-estimating π can change decisions only at $\lambda > 0$. (ii) Errors of each factor are weighted by the other:

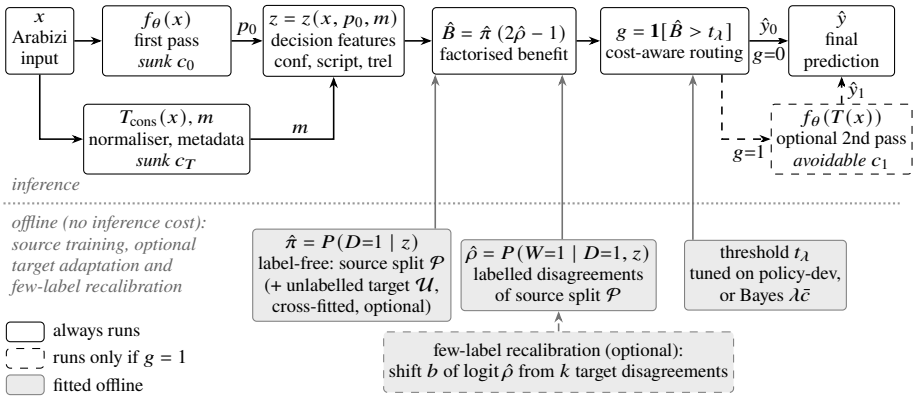


Fig. 1. SelectNorm. Upper lane (inference): $f_\theta(x)$ and $T_{\text{cons}}(x)$ always run, so c_0 and c_T are incurred before the decision g; the dashed second pass $f_\theta(T(x))$ is the only avoidable cost and runs only when $g = 1$. Lower lane (grey, offline): $\hat{\pi}$ (label-free, optionally adapted on unlabelled target text), $\hat{\rho}$ (source disagreements), the threshold t_λ and the optional few-label shift of $\hat{\rho}$.

Proposition 1 (Plug-in regret). *With $R(g)$ the risk in Eq. (1), $R(\hat{g}) - R(g^\star) \le \mathbb{E}\big[\,|2\rho - 1|\,|\hat{\pi} - \pi| + 2\hat{\pi}\,|\hat{\rho} - \rho|\,\big]$.*

This standard plug-in bound (Appendix A) motivates measuring the two estimation errors separately. Exactness of Eq. (3) requires binary labels and the 0-1 loss; macro-F1 is not a sum of per-input losses, so the rule is accuracy-optimal and only approximately macro-F1-optimal.

Adaptation to a new dialect. On a target distribution Q, π_Q can be re-estimated without labels; transfer of $\hat{\rho}$ assumes $P_Q(W \mid D{=}1, z) = P(W \mid D{=}1, z)$, which we test rather than assume. When ρ shifts, we model $\rho_Q(z) = \sigma\big(\operatorname{logit}\rho(z) + b\big)$ and estimate b from k labelled target disagreements.

Proposition 2 (Label efficiency of disagreement-targeted labelling). *Under this model, with π_Q known, the Fisher information about b of k target labels drawn at random is $k\,P_Q(D{=}1)\,\mathbb{E}_Q[\rho_Q(1-\rho_Q) \mid D{=}1]$, and of k labels drawn among disagreements $k\,\mathbb{E}_Q[\rho_Q(1-\rho_Q) \mid D{=}1]$; targeted labelling is $1/P_Q(D{=}1)$ times more informative.*

It concerns estimating b, not the accuracy of the resulting router.

Information access. The policy never reads p_1 at decision time; p_1 is used only on the policy split and on unlabelled target text disjoint from the evaluated fold. The oracle $g^{\text{or}} = \mathbf{1}[e_0{=}1 \wedge e_1{=}0]$ reads labels.

4 SelectNorm

Module A: conservative normaliser. T changes only Latin-script tokens (Algorithm 1). A token is looked up in a lexicon of Arabizi→CODA pairs built from 80% of TArC [14];

Algorithm 1 Conservative normalisation T_{cons}

Require: sentence x, lexicon $\mathcal{L}$, rules R, frequency lists F
Ensure: $T(x)$, metadata m
$\tau \leftarrow$ tokens of x; $\kappa \leftarrow$ share of foreign tokens in τ by F vs. $\mathcal{L}$
for each Latin-script token $t \in \tau$ **do**
  if t is laughter **then** emit هههه
  else if t is foreign (if short or function-like: iff $\kappa > 1/2$) **then** keep t
  else if t is a listed negator **then** emit its fixed form
  else if $t \in \mathcal{L}$ **then** emit $\mathcal{L}(t)$
  else if t matches *ma…ch* **then** emit ما R(stem) ش
  else emit $R(t)$ and count ambiguous characters
  update m with the branch taken
return joined output, m

unseen tokens fall back to Tunisian character rules (*3*→ع, *7*→ح, *9*→ق, *5*→خ, *ch*→ش, word-initial vowels to alef, medial short vowels dropped). Three safeguards define T_{cons}: (i) *protection* of French and English tokens, decided from word frequencies against TArC evidence, with short or function-like tokens (*la*, *ma*, *el*) read in the language of their context; (ii) *negation*: frequent negators (*mouch*, *mech*) are mapped by a fixed list and the circumfix *ma…ch* is split as ما + stem + ش; (iii) *non-lexical cues*: laughter (*hhhh*, *mdr*) becomes هههه. T_{rule} (rules only) and T_{lex} (no safeguards) are controls. Each call also returns token-level metadata m (counts per branch, ambiguous characters).

Module B: benefit estimation. The decision-point features z form three groups: *conf* (maximum probability, margin and entropy of the temperature-scaled p_0), *script* (length, Latin share, digits used as letters, emoji and upper-case rates) and *trel* (the metadata m as rates per Latin token, and the share of changed tokens). $\hat{\pi}$ and $\hat{\rho}$ are logistic regressions on standardised z (gradient boosting as ablation); $\hat{\rho}$ is fitted on the disagreements of the policy split only (Algorithm 2).

Module C: cost-aware routing. At inference (Algorithm 3) $\hat{B}(z) = \hat{\pi}(z)(2\hat{\rho}(z) - 1)$ is compared with a threshold t_λ: either the Bayes threshold $\lambda\bar{c}$ of Eq. (2), or, as for every baseline, a threshold tuned for each λ on the policy development split. No threshold is tuned on target data. With encoder cost C_f, the expected inference cost is $C_f + C_T + O(|z|) + \mathbb{E}[g]\, C_f$; only the last term is avoidable.

5 Experimental Setup

Data and classifiers. f is trained on Arabic-script TSAC [22]; the human Latin-script TSAC training comments, never seen by f, form the policy pool (Table 1). Test sets cover the same corpus (S_3), another Tunisian corpus and platform (S_4, TUNIZI [10]), Algerian (S_5, NArabizi [29,33]) and Moroccan Arabizi (S_6, MYC [17]). Mixed-script sentences (minority alphabet > 20%) are excluded; corpora were cloned at fixed commits and are not redistributed. f_θ is a logistic regression on mean-pooled frozen representations of

Algorithm 2 Training

Require: f, T, labelled policy split $\mathcal{P}$; optional: unlabelled target fold $\mathcal{U}$; k labelled target disagreements $\mathcal{K}$
Ensure: $\hat{\pi}$, $\hat{\rho}$ (and shift $\hat{b}$)
for $x \in \mathcal{P} \cup \mathcal{U}$ **do** compute p_0, p_1, z, D
$\hat{\rho} \leftarrow$ fit $P(W{=}1 \mid z)$ on $\{i \in \mathcal{P} : D_i{=}1\}$
$\hat{\pi} \leftarrow$ fit $P(D{=}1 \mid z)$ on $\mathcal{P}$
    (or on $\mathcal{P} \cup \mathcal{U}$, equal total weight)
if $\mathcal{K} \neq \emptyset$ **then** ▷ few-label shift
    $\hat{b} \leftarrow \arg\max_b \ell(b)$, with $r_i = \operatorname{logit} \hat{\rho}(z_i)$:
    $\ell(b) = \sum_{\mathcal{K}} \log \mathrm{Bern}(W_i; \sigma(r_i{+}b)) - b^2/2$

Algorithm 3 Inference

Require: x, f, T, $\hat{\pi}$, $\hat{\rho}$, threshold t_λ ($\lambda\bar{c}$ or tuned on policy-dev)
Ensure: label $\hat{y}$, action g, score $\hat{B}$
$p_0 \leftarrow f(x)$; $(x', m) \leftarrow T(x)$ ▷ sunk
$z \leftarrow z(x, p_0, m)$
$\hat{B} \leftarrow \hat{\pi}(z)(2\hat{\rho}(z) - 1)$
$g \leftarrow \mathbf{1}[\hat{B} > t_\lambda]$
if $g = 1$ **then** $\hat{y} \leftarrow \mathbf{1}[f(x') > 1/2]$
else $\hat{y} \leftarrow \mathbf{1}[p_0 > 1/2]$
return $\hat{y}, g, \hat{B}$

Table 1. Data after de-duplication and near-copy removal (character 3-gram cosine ≥ 0.9 against every training or policy sentence). S_5^{AR}: human Arabic-script transliteration of S_5 [33].

Set	Variety, script	N	Pos.%	Role
TSAC-AR	Tunisian, Arabic	5,564	45.0	train f (90/10)
TSAC-LAT train	Tunisian, Latin	6,204	57.3	policy pool
S_3 TSAC-LAT test	Tunisian, Latin	1,425	55.2	test
S_4 TUNIZI	Tunisian, Latin	2,875	48.8	test
S_5 NArabizi	Algerian, Latin	729	52.0	test
S_5^{AR} NArabizi gold	Algerian, Arabic	729	52.0	RQ4
S_6 MYC-LAT	Moroccan, Latin	4,191	66.4	test

XLM-R base [5], mE5 base [34], LaBSE [9] or MARBERTv2 [1] at pinned revisions, with regularisation and a temperature [15] fitted on the 10% Arabic-script development split. No encoder is fine-tuned and no Latin-script sentence trains f. In an exploratory regime (+SYN), logged as a deviation before it was run, f also sees a synthetic Latin rendering of each training sentence.

Systems. B0 keeps x; B1 always normalises with T_{cons}; B2 routes at random at SelectNorm's selection rate (20 draws); B3 thresholds the confidence of p_0; B4 estimates $B(z)$ directly as $P(\mathrm{help} \mid z) - P(\mathrm{harm} \mid z)$, a post-hoc deferral rule [18]; B4b predicts each view's error [20]; B6 is the oracle; R averages both views' probabilities. SelectNorm-src uses source estimates; SelectNorm-adapt, the primary system, re-estimates $\hat{\pi}$ on unlabelled target text with two-fold cross-fitting by source sentence. All thresholds are tuned per λ on the policy development split (the Bayes threshold $\lambda\bar{c}$ is a variant); nothing is tuned on a test set.

Few-label setting. Each test set is split into two folds by source sentence; to predict one fold, $k \in \{8, \ldots, 128\}$ labels are drawn from the other, at random or among its disagreements, and re-estimate b (SelectNorm), re-fit B4, re-tune B3, or choose B0 or

B1 for the fold (set-level choice). A replay of all 8,000 draws, checked against indices logged by a rerun, found no sentence whose own label informed its prediction. On S_5 at k=128 only 97–116 disagreements exist per fold, so 375 targeted draws are smaller than k. Each k uses 5 draws × 5 seeds and the $\lambda = 0$ threshold.

Mechanisms, statistics and costs. We measure (M1) the AUC of $\hat{\pi}$ for disagreements; (M2) the AUC of $\hat{\rho}$ for the winning view among disagreements, its calibration gap, and a ceiling from 5-fold cross-validation of $\hat{\rho}$ on *target* labels; (M3) the gain over the best view of the rule $\hat{\rho} > 1/2$; (M4) the oracle headroom, against the headroom between two classifiers trained on bootstrap resamples and applied to the same input. Intervals are 95% percentile intervals from 1,000 seed-averaged bootstrap resamples of test sentences, without refitting. Macro-F1 is primary; results average five seeds, which fix the policy split and random draws. Differences are tested with a paired bootstrap over source sentences [21,8] (10,000 resamples) with Holm correction over the 20 main comparisons per encoder. A difference is *relevant* if it is at least one point and significant after correction; the protocol's wording requires only the unadjusted interval to exclude zero, which coincides in the primary regime (for +SYN we report both). Non-significance is not read as equivalence. Costs are relative to one pass of f on x: c_T is measured; c_0, c_1 are *estimated* by a per-encoder model $a + b \cdot$ tokens fitted to measured batch-1 CPU latencies and checked by end-to-end timing.

Status of analyses. Pre-specified: main comparisons and statistics, the λ grid, the embedding, boosting and T_{lex} ablations, the few-label setting and RQ4. Exploratory, logged after the first mE5 results: T_{cov}, +SYN and M1–M4. Added in revision: mechanism intervals, sensitivity and equivalence readings, end-to-end timing, the normaliser audit and the input-dependent cost threshold.

6 Results

6.1 Selective normalisation against strong baselines (RQ1, RQ2)

Always normalising (B1) improves over the original input on the Tunisian and Algerian sets for every encoder, by 2.3–18.6 points (Table 2); on MYC it gains 8.8 points with XLM-R and loses 7.9 with LaBSE. Learned routing adds no consistent improvement. Against the direct-gain router B4 no difference is significant, and every 95% interval lies inside $[-0.63, +0.58]$, within the 1-point margin in all 16 cells (a post hoc equivalence reading; Table S1); SelectNorm-adapt and -src differ by −0.25 to +0.54 (three small differences significant). Against B1 and B3, SelectNorm-adapt is relevantly better in one cell (LaBSE, MYC: +5.5, +5.0), where normalising harms and thresholding fails to stop it, and relevantly worse in one (XLM-R, MYC: −3.8), where the router keeps the original view for about 40% of inputs although normalising is right; in the LaBSE MYC cell it is also relevantly worse than B0, the better view there (−2.5). Elsewhere only 4 (B1) and 6 (B3) of 16 intervals lie within ±1, and the other non-significant ones span −1.3 to +2.4, so small gains or losses cannot be excluded.

Table 2. Macro-F1 ($\lambda = 0$, mean of five seeds), classifier trained on Arabic script only. Bold: best deployable system per column; shaded: SelectNorm-adapt (primary). $^{\triangledown}$/$^{\blacktriangle}$: baseline significantly worse/better than SelectNorm-adapt (paired bootstrap, Holm-corrected $p < 0.05$ over 20 comparisons per encoder; tested for B0, B1, B3, B4, SelectNorm-src). †Two passes; ‡uses test labels.

	mE5				**XLM-R**			
	S_3	S_4	S_5	S_6	S_3	S_4	S_5	S_6
B0 original	63.4$^{\triangledown}$	61.5$^{\triangledown}$	63.6$^{\triangledown}$	**54.1**	54.5$^{\triangledown}$	48.8$^{\triangledown}$	48.6$^{\triangledown}$	63.5$^{\triangledown}$
B1 always normalise	**82.0**	**67.2**	69.4	52.9	72.8	**64.5**	63.1	72.3$^{\blacktriangle}$
B2 random, matched rate	80.0	66.5	68.5	53.2	70.2	63.3	61.9	68.9
B3 confidence threshold	82.0	67.2	69.4	52.9	72.9	64.4	63.0	**72.3**$^{\blacktriangle}$
B4 direct gain	81.4	67.0	69.9	53.6	73.5	63.9	63.3	68.7
B4b per-view loss	81.6	67.1	69.9	53.7	**73.9**	63.9	**63.4**	68.2
SelectNorm-src	81.3	67.0	**70.0**	53.7	73.8	63.8	63.4$^{\blacktriangle}$	68.0$^{\triangledown}$
SelectNorm-adapt	81.3	66.9	69.8	53.7	73.8	63.8	63.1	68.5
Bayes threshold $\lambda\bar{c}$	81.4	66.9	69.6	53.5	73.9	64.0	63.4	69.0
R two-view ensemble†	78.9	66.5	69.7	54.4	67.3	57.2	54.7	69.7
B6 oracle‡	90.0	81.3	81.8	63.8	83.1	75.2	70.4	82.1

	LaBSE				**MARBERTv2**			
	S_3	S_4	S_5	S_6	S_3	S_4	S_5	S_6
B0 original	69.6$^{\triangledown}$	61.1$^{\triangledown}$	65.6$^{\triangledown}$	**63.4**$^{\blacktriangle}$	64.3$^{\triangledown}$	62.4$^{\triangledown}$	60.0	52.5$^{\triangledown}$
B1 always normalise	83.6	**69.1**	70.9	55.5$^{\triangledown}$	82.1	67.7	62.3	58.3
B2 random, matched rate	81.0	67.3	69.5	60.4	79.9	67.3	61.9	56.7
B3 confidence threshold	83.7	69.1	70.9	55.9$^{\triangledown}$	**82.3**	68.1	62.3	**58.3**
B4 direct gain	83.4	68.6	71.6	60.6	82.1	68.2	**63.0**	57.6
B4b per-view loss	83.3	68.9	**71.6**	60.5	82.1	68.1	62.8	57.6
SelectNorm-src	83.4	68.7	71.1	60.4$^{\triangledown}$	82.2	68.3	62.9	57.7
SelectNorm-adapt	83.4	68.8	71.5	60.9	82.1	**68.3**	62.9	57.7
Bayes threshold $\lambda\bar{c}$	**83.7**	68.8	71.3	60.8	82.2	68.1	62.5	57.9
R two-view ensemble†	81.7	68.4	72.1	60.5	82.3	69.7	62.4	58.4
B6 oracle‡	91.5	82.2	85.7	75.2	93.6	84.2	77.6	70.4

6.2 Accuracy and cost (RQ3)

B0 costs 1.00 and B1 0.97–1.04 passes on x (the normaliser adds about 0.4%). At $\lambda = 0$ SelectNorm-adapt costs 1.69–1.88 passes for mE5 and 1.43–1.88 across encoders, against 1.89–2.05 for the ensemble; raising λ lowers the cost (mE5: 1.14–1.45 at $\lambda = 0.2$) while macro-F1 falls towards B0 (Fig. 2). Routing is never cheaper than the better single view and never relevantly more accurate (at most +0.6, not significant); it saves compute only relative to running both views. End-to-end timing of the deployed pipeline (200 sentences per set; live decisions reproduce all 184,400 archived ones) agrees: B0 22–31 ms; B1 0.91–1.05, SelectNorm-adapt 1.31–1.97, R 1.91–2.07 passes. Thresholding with each input's own cost $\lambda c_1(x)$ (exploratory) changes macro-F1 by −2.0 to +0.6 and cost by 0 to −0.11 passes, moving operating points along the same frontier (no worse on both in 29 of 80 cells, trading accuracy for cost in 48).

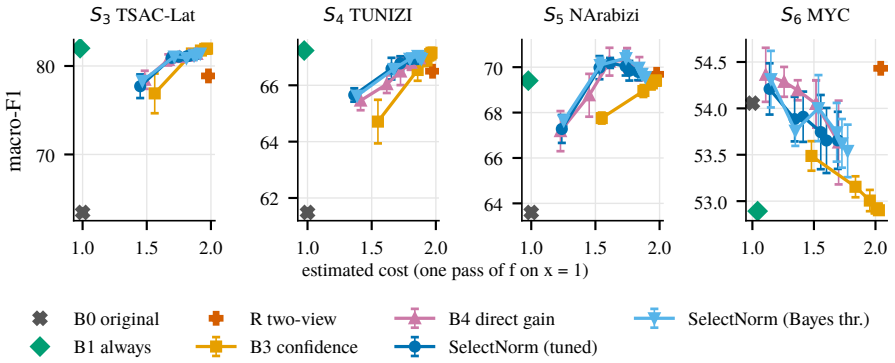


Fig. 2. Macro-F1 against estimated cost (mE5, batch size 1, CPU; one pass of f on $x = 1$) for $\lambda \in \{0, .01, .02, .05, .1, .2\}$; bars: s.d. over seeds. Routers and B3 are charged the first pass and the normaliser (sunk) plus the second pass when selected; B1 the normaliser and one pass; R both passes.

Table 3. Mechanisms (T_{cons}, Arabic-script f, five seeds). M1: disagreement rate (%), AUC of target-cross-fitted $\hat{\pi}$. M2: win rate of $T(x)$ on disagreements (%), AUC of source and of target-CV $\hat{\rho}$ (not deployable), calibration gap of source $\hat{\rho}$. M3: gain over the best view of $\hat{\rho} > 1/2$. M4: oracle headroom, and the same minus the bootstrap-classifier reference. °: AUC interval contains 0.5; *: interval excludes 0 (95%; Table S4).

		(M1) disagreement		(M2) direction				(M3) benefit		(M4) headroom	
Enc.	Set	$P(D)$	AUC_π	$P(W\|D)$	AUC_ρ	$\text{AUC}_\rho^{\text{tgt}}$	gap	Δ_{src}	Δ_{tgt}	oracle	vs. ref.
mE5	S_3	34	0.68	77	0.66	0.64	+3	-0.6*	+0.0	8.0	-0.3
	S_4	33	0.64	57	0.56	0.58	+19*	-0.3	-0.1	14.1	+4.6
	S_5	29	0.62	58	0.63	0.57°	+15*	+0.2	+0.7	12.4	+3.2
	S_6	21	0.74	47	0.54	0.56	+20*	-0.5	+0.2	9.7	+4.6
XLM-R	S_3	36	0.64	72	0.71	0.68	+3	+1.0*	+0.8	10.3	+5.9
	S_4	31	0.66	64	0.57	0.57	+11*	-0.5	+0.3	10.7	+7.6
	S_5	23	0.69	69	0.45°	0.60°	+11*	+0.3	-0.7	7.3	+2.1
	S_6	28	0.72	69	0.44	0.66	-8*	-3.3*	+1.1*	9.8	+4.5
LaBSE	S_3	30	0.69	73	0.61	0.53°	-0	+0.1	-0.1	7.9	+1.3
	S_4	34	0.65	62	0.57	0.59	+6*	-0.4	-0.4	13.0	+5.7
	S_5	34	0.65	58	0.59	0.55°	+3	+0.4	+0.2	14.9	+7.5
	S_6	31	0.68	33	0.57	0.64	+16*	-2.6*	+0.8*	11.8	+5.3
MARBERTv2	S_3	39	0.69	70	0.63	0.60	+1	+0.1	-0.3	11.5	+0.8
	S_4	37	0.71	60	0.61	0.64	+13*	+0.4	+2.8*	16.5	+2.0
	S_5	31	0.73	57	0.63	0.54°	+13*	+0.2	+1.1	15.3	+3.6
	S_6	30	0.71	59	0.55	0.63	+4*	-0.3	+1.1*	12.1	-0.7

6.3 Where per-input routing fails (RQ5)

(M1) Disagreement is moderately predictable. 21–39% of inputs change label under normalisation; the label-free estimate ranks them with AUC 0.62–0.74, all intervals

within $[0.57, 0.77]$ (Table 3, Fig. 3). Target re-estimation helps most on MYC (mE5: 0.67 to 0.74), but at $\lambda = 0$ the rule depends on $\hat{\rho}$ alone.

(M2) Direction is weakly predictable from z, even in domain. Source $\hat{\rho}$ reaches AUC 0.54–0.71 in 14 of 16 cells (upper bounds ≤ 0.76); for XLM-R the interval includes 0.5 on Algerian text and lies below it on Moroccan text ($[0.41, 0.48]$). Fitted on target labels by cross-validation, a reference no deployable system has, $\hat{\rho}$ reaches 0.53–0.68 (upper bounds ≤ 0.73; five intervals include 0.5). With linear estimators the features carry little information about direction in any domain tested; better features or estimators are not excluded. What changes across domains is the *level* of ρ: the win rate ranges from 33% to 77%, and source $\hat{\rho}$ is miscalibrated out of domain (gaps up to +20 points; intervals exclude zero in 11 of 12 cells).

(M3, M4) Benefit and headroom. The rule $\hat{\rho} > 1/2$ gains −3.3 to +1.0 points over the best view with source estimates (one interval above zero, three below) and −0.7 to +2.8 with the target-CV ceiling (upper bounds $\leq +3.9$): most of the gap to the oracle remains when the router is fitted on the target. The oracle gains 7–17 points over the best view, but two classifiers differing only in a bootstrap resample of their training data show 3–15 points of such headroom on the same input; this reference depends on how the classifiers are made (2–6 with 80% subsamples, 5–22 with weaker regularisation; Table S5) and is not a decomposition.

Linguistic conditions. On Tunisian and Algerian sets, always normalising helps less on inputs with many French or English tokens or low lexicon coverage, for every encoder except XLM-R on TUNIZI (French/English); negation and length show no consistent pattern. On MYC every condition changes sign with the encoder (Table S6).

6.4 Few target labels (RQ6), normaliser and classifier regime

With k labelled target sentences, re-estimating the shift of $\hat{\rho}$ changes macro-F1 relative to zero-shot SelectNorm-adapt by −1.4 to +1.5 points across encoders, regimes, sets and k (mean −0.01). Although targeted labels are $1/P(D{=}1)$ times more informative about the shift (Proposition 2), targeted and random labels give results within −0.1 to +0.2 points of each other at every k (averaged, descriptive). In the Arabic-script regime SelectNorm beats a set-level choice between B0 and B1 when $k \leq 16$ (by 2.6–2.9 points on average); the advantage shrinks to 0.3 points at $k = 128$, consistent with M2.

The safeguards matter more than routing: always normalising with T_{lex} or T_{rule} loses 4–11 points on every set for mE5, and routing over them recovers only part of the loss (Table S7). Keeping out-of-lexicon tokens in Latin script (T_{cov}) reduces disagreements by about a third; it is worse than T_{cons} on the Tunisian sets and slightly better on S_5 and S_6. On NArabizi (RQ4), B1 agrees with f's prediction on the human transliteration for 73–77% of inputs, against 61–67% for B0 and 73–76% for SelectNorm-adapt; f reaches 65–77 macro-F1 on the human transliterations. In the +SYN regime (Table S2) the original view improves, the win rate on disagreements moves towards 50%, and SelectNorm-adapt is relevantly better than the better view once with Holm correction (MARBERTv2, S_4, +1.1) and twice more under the unadjusted criterion (mE5, S_3 +2.1,

Table 4. Test sentences with archived predictions (mE5, $\lambda = 0$, SelectNorm-adapt): x above $T_{\text{cons}}(x)$; gold y (1 = positive); posteriors p_0, p_1; routing g and prediction $\hat{y}$ for seeds 0–4. Sources: TSAC test-file line (LGPL-3.0); NArabizi `sent_id` (UD release, CC BY-SA 4.0); MYC row of `DATA_CLEANED.csv`, text withheld (no licence).

Source, effect	Original x / normalised $T_{\text{cons}}(x)$	y	p_0	p_1	seeds 0–4
TSAC neg. 796	*mech 5arej 3lik el role*	0	.565	.127	g 11111
help	role مش خارج عليك ال				$\hat{y}$ **00000**
TSAC pos. 86	*Metrobi mouch kif ijale4 la5er*	1	.806	.004	g 11111
harm	مترﺑﻲ موش كيف يجلذ لخر				$\hat{y}$ **00000**
NArabizi 2	*salem 3alikoum inchalah le pondium et les midailes d'or*	1	.175	.842	g 11111
help	سلام عليكوم ان شاء الله لا بنديوم ات لس ميديلس د ' اور				$\hat{y}$ 11111
NArabizi 4609	*oui le Maroc aktar jahiziya li had tasfiyat la 3ala lmostawa tandimi*	1	.519	.088	g 11111
harm	وي لا Maroc اكتر جهيزيا اللي حد تسفييت لا علي لمستوا تنديمي				$\hat{y}$ **00000**
MYC 9079	[withheld] two *ma. . . ch* negations split as ما . . . ش,	1	.645	.250	g **00**111
harm (s2–4)	letter elongations squeezed, emoji kept				$\hat{y}$ **11000**
MYC 14543	[withheld] two Latin-script loanwords protected,	1	.480	.302	g **00001**
both wrong	negated verb split, emoji kept				$\hat{y}$ **00000**

S_4 +1.4), and relevantly worse in two MYC cells under both (LaBSE −2.4, MARBERTv2 −1.8); it is never significantly better than B4 (worse once, −0.4).

6.5 Qualitative examples

Table 4 shows six hand-picked (not representative) test sentences. In TSAC neg. 796 the negator *mech* is mapped by the fixed list and French *role* is protected; in TSAC pos. 86, *ijale4* and *la5er* fall back to character rules. In both NArabizi sentences the French article *le* is read as Arabizi (the context is mostly Arabizi) and mapped by the negator list to لا (*no*), yet the normalised view is right in sentence 2 and wrong in 4609. The router selects the normalised view for all four in every seed: its features do not separate these cases. In MYC 9079 the decision varies with the policy split.

7 Discussion

Findings. Transfer matters in specific cells: for XLM-R on Algerian and Moroccan text, source $\hat{\rho}$ ranks direction at or below chance, and fitting it on the target turns a loss of up to 3.3 points into a small gain. Elsewhere the limit is in-domain, and richer variants did not change this (16 embedding components: −1.2 to +1.1, no consistent sign; gradient boosting, mE5 only: −0.6 to +0.7). The win rate on disagreements varies far more between encoder–dialect pairs (33–77%) than our features resolve within a set, and with 64–128 target labels a set-level choice is on average within 0.3–0.7 points of

SelectNorm with the same labels (descriptive). The factorisation adds a diagnosis, not an accuracy gain.

Explanations. Our reading is that decision-point features say little about which view is right, and that much of the oracle headroom is of the size classifier variability alone produces (M4). These are limits of the features and estimators evaluated; other features, fine-tuned representations or estimators might do better, and the results do not show that the conditional advantage is intrinsically unpredictable. Targeted labels estimate the shift more efficiently but cannot repair a weak ranking. Transliteration is known to help unseen scripts on average [25,23]; on Moroccan text its sign changes with the encoder. In practice, safeguards matter more than routing, and normalising is best decided per corpus or dialect.

8 Limitations

Task and models. All experiments use binary sentiment, where the factorisation is exact with ρ alone; the K-class form needs two conditional win rates. Encoders are frozen and heads linear, and all conclusions are restricted to this setting; fine-tuning could change how often normalisation helps and how predictable the winning view is. We did not fine-tune (CPU only); +SYN is only a partial probe. *Normaliser.* T is built from Tunisian resources (TArC lexicon, Tunisian digit conventions such as $8 =$ ٥). Its safeguards were tuned on the TArC held-out split and no independent transliteration test set was available, so its intrinsic accuracy is *not independently validated*; it is lower on Algerian NArabizi, and with no Moroccan transliteration gold its validity on MYC is unmeasured. *Data.* Cross-corpus differences mix dialect with platform, topic, period and label distribution; we do not attribute them to dialect alone. TUNIZI, the NArabizi sentiment layer and MYC state no licence, and MARBERTv2 has none on its model card; we use them for research only and do not redistribute them. *Development.* The pipeline was first exercised with a character n-gram classifier whose test results we inspected; this motivated the few-label setting. *Analysis.* Mechanism intervals cover test-sentence resampling, not estimator refitting; the target-label ceiling uses linear $\hat{\rho}$ on the same features. Features computed after $f(T(x))$ were not studied, as they need the second pass on every input. Equivalence statements are post hoc and not corrected for multiplicity. The qualitative examples are hand-picked. *Costs.* Costs are estimated from a token-length model and checked by timing 200 sentences per set on one 4-core CPU; relative costs on accelerators differ, especially for short inputs.

9 Conclusion

We studied per-input normalisation of Arabizi for frozen encoders with linear heads, using the disagreement factorisation of the expected benefit as an analysis device. Normalisation improves macro-F1 on Tunisian and Algerian Arabizi, and its effect on Moroccan text depends on the encoder. Learned per-input routing, including SelectNorm, does not consistently exceed the better single view or strong baselines, and it costs more than either single view because the first pass is sunk. Measured separately, disagreement is

moderately predictable from decision-point features, whereas the conditional advantage of the normalised view is weakly predictable under the features and estimators evaluated, also when fitted on target labels. The usable signal is mostly set-level, which about a hundred target labels recover.

Code and Reproducibility. Code, configuration, the pre-registered protocol, logs, per-example predictions, supplementary Tables S1–S8 and all figure data are at [GITHUB_REPOSITORY_URL], with dependencies, commands, pinned encoder revisions and corpus commits, and instructions for obtaining the corpora, which are not redistributed. A script recomputes every test of Table 2 from the exports. Quoted NArabizi sentences are from the UD release https://github.com/UniversalDependencies/UD_Maghrebi_Arabic_French-Arabizi (CC BY-SA 4.0); quoted TSAC sentences are from https://github.com/fbougares/TSAC (LGPL-3.0).

Disclosure of Interests. [To be confirmed by the authors.] The authors have no competing interests to declare that are relevant to the content of this article.

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A Proofs

Bayes rule and factorisation. For fixed z, the conditional risk of $g(z) = a$ is $\mathbb{E}[e_a \mid z] + \lambda\,\mathbb{E}[c_0 + c_T \mid z] + \lambda\bar{c}(z)\,a$; the middle term does not depend on a, so $a = 1$ is strictly better iff $B(z) > \lambda\bar{c}(z)$, and pointwise minimisation minimises Eq. (1). If $D = 0$, $e_0 = e_1$; if $D = 1$ and $y \in \{0, 1\}$, $e_0 = W$ and $e_1 = 1 - W$. Hence $e_0 - e_1 = D(2W - 1)$ and $B(z) = P(D{=}1 \mid z)\big(2P(W{=}1 \mid D{=}1, z) - 1\big)$; for K classes $e_0 - e_1 = D\big(\mathbf{1}[\hat{y}_1{=}y] - \mathbf{1}[\hat{y}_0{=}y]\big)$ gives $\pi(\rho_1 - \rho_0)$. □

Proposition 1. Where $\hat{g}(z) \neq g^\star(z)$, the conditional excess risk is $|B - \lambda\bar{c}| \leq |\hat{B} - B|$, because $\hat{B} - \lambda\bar{c}$ and $B - \lambda\bar{c}$ differ in sign (or $\hat{B} = \lambda\bar{c}$). Hence $R(\hat{g}) - R(g^\star) \leq \mathbb{E}|\hat{B} - B|$, and $\hat{B} - B = (\hat{\pi} - \pi)(2\rho - 1) + 2\hat{\pi}(\hat{\rho} - \rho)$ with the triangle inequality concludes. □

Proposition 2. With $q_i = \sigma(\operatorname{logit}\hat{\rho}(z_i) + b)$, a labelled target example contributes the log-likelihood $D_i[W_i \log q_i + (1{-}W_i)\log(1{-}q_i)]$ (if $D_i = 0, e_0 = e_1$ whatever y_i), hence Fisher information $D_i\,q_i(1{-}q_i)$ about b. Its expectation is $P_Q(D{=}1)\,\mathbb{E}_Q[q(1{-}q) \mid D{=}1]$ under random sampling and $\mathbb{E}_Q[q(1{-}q) \mid D{=}1]$ among disagreements (sampled without labels, as D is observable); information adds over draws. □

B Disagreement–Advantage Diagnostic Map

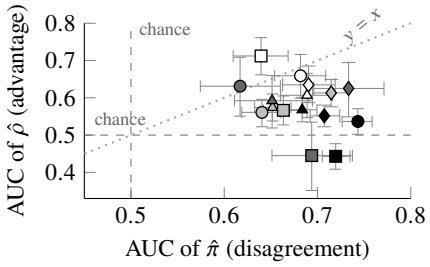


Fig. 3. Disagreement–advantage diagnostic map (data of Table 3; bars: 95% intervals). Marker: ○ mE5, □ XLM-R, △ LaBSE, ◇ MARBERTv2; fill: □ S_3, ▫ S_4, ▪ S_5, ■ S_6. x: AUC of target-cross-fitted $\hat{\pi}$ (M1); y: AUC of source $\hat{\rho}$ (M2); dashed: chance; dotted: equal AUC. 14 of 16 points lie below the diagonal; the two near 0.44 are XLM-R on S_5 and S_6. Linear estimators on z only.